

NANYANG TECHNOLOGICAL UNIVERSITY

Lecture 11

EE3010: Electrical Devices and Machines

School of Electrical and Electronic Engineering

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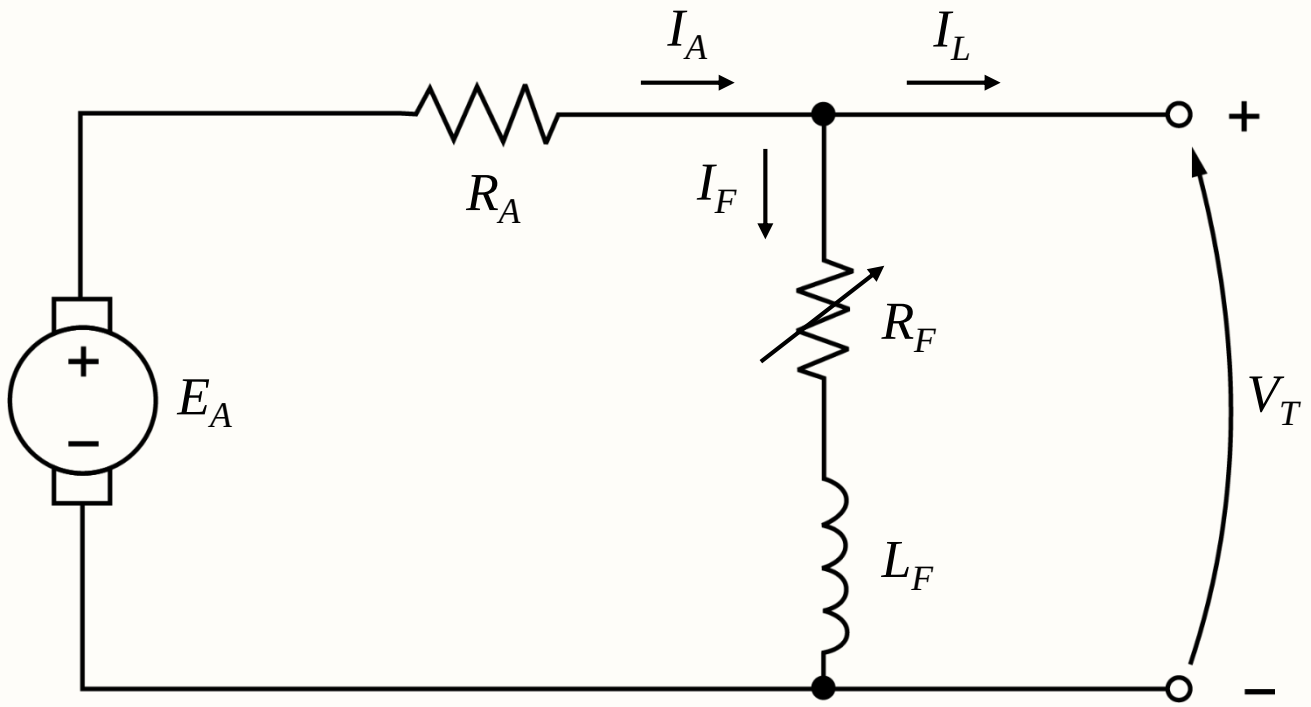
Learning Objectives

By the end of this lecture, you should be able to:

- Describe the voltage build-up in a shunt DC generator.
- Interpret the meaning of critical resistance and how to obtain it.
- Describe the voltage-current characteristics of a DC shunt generator.
- Carry out graphical analysis of a DC shunt generator.
- Describe the losses and power flows, and calculate the generator efficiency at given operating condition.

The Shunt DC Generator

- The generator supplies its own field current, by having its field winding connected directly across the terminals of the machine.



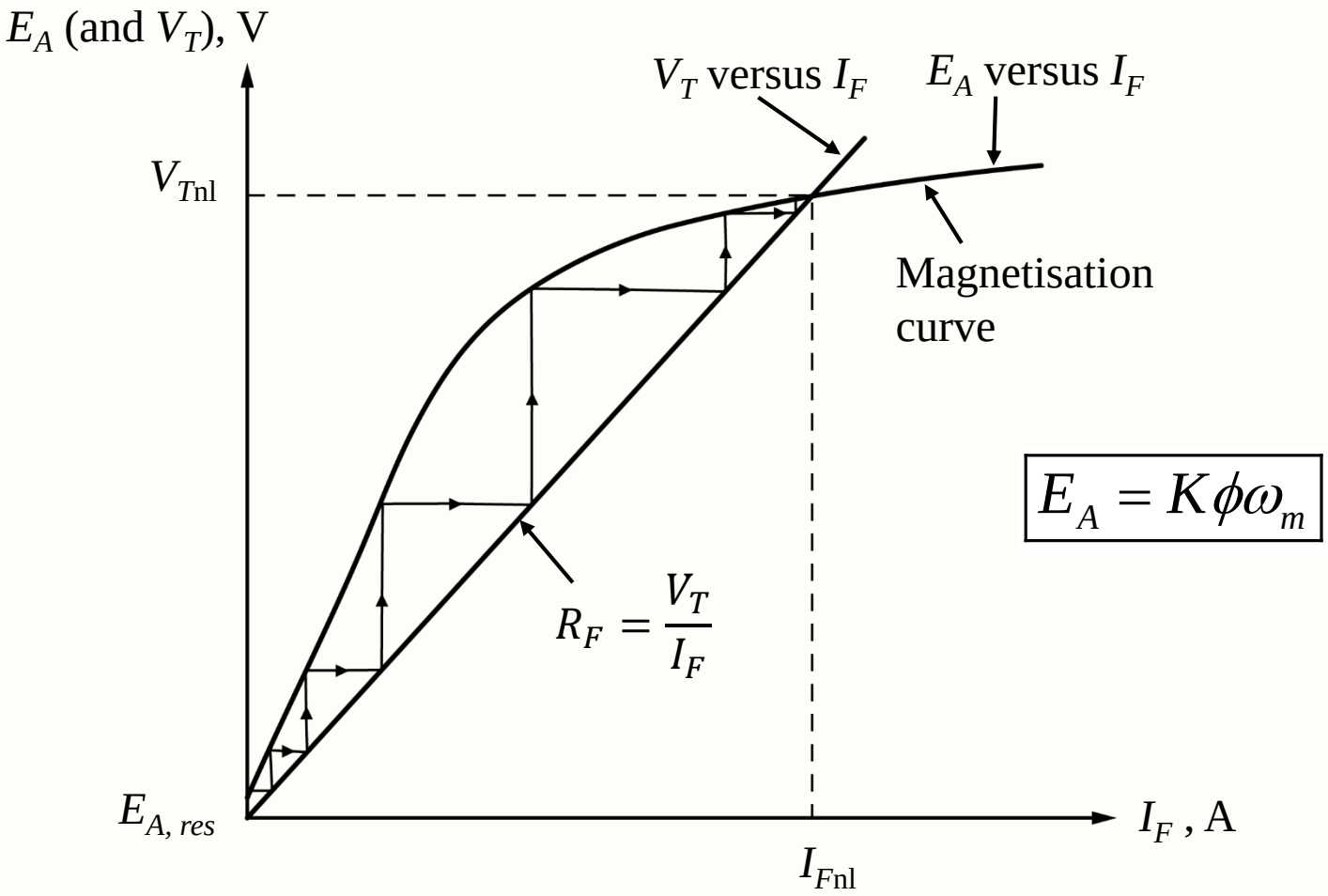
Equivalent circuit of a shunt DC generator.

$$I_A = I_F + I_L,$$
$$V_T = E_A - I_A R_A,$$
$$I_F = \frac{V_T}{R_F}$$

The Shunt DC Generator

- The armature current of the machine supplies both the field circuit and the load attached to the machine.
- This type of generator has an advantage over the separately excited generator, in that no external power supply is required for the field circuit.
- If the generator supplies its own field current, how does it get the initial field flux when it is first turned on?

Voltage Build-up in a Shunt DC Generator



Voltage build-up on starting in a shunt DC generator.

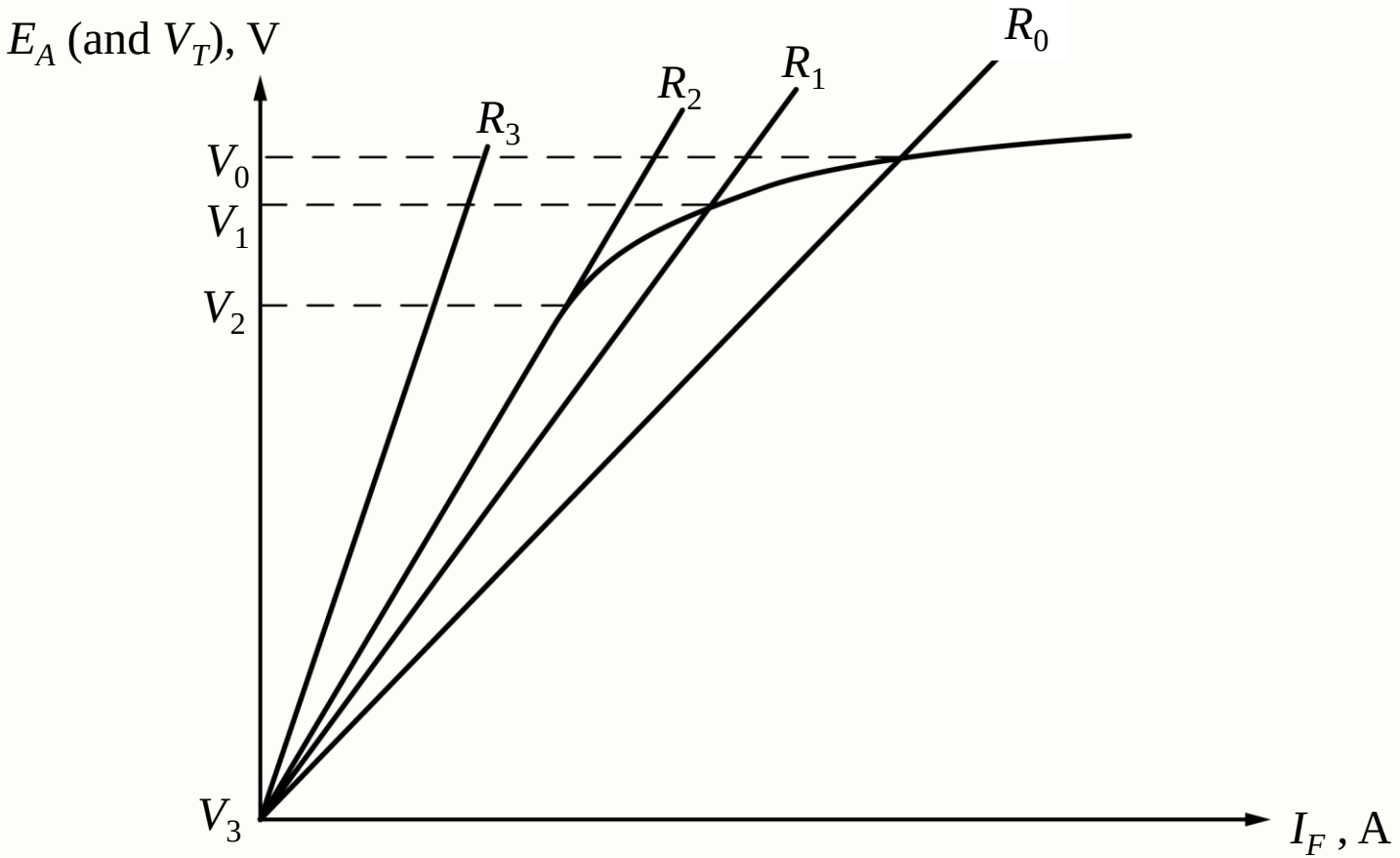
Voltage Build-up in a Shunt DC Generator

- The voltage build-up depends on the presence of a residual flux ϕ_{res} in the machine. When the generator first starts to turn with no load connected to it, an internal voltage given by $E_A = K\phi_{res}\omega_m$ will be generated. A small V_T appears across the field windings and I_F flows which produces an mmf, and the flux increases which in turn increases E_A, etc.
- It is the effect of the magnetic saturation in the machine which eventually limits the terminal voltage of the generator.
- The field resistance should be less than $R_{critical}$ for voltage build-up to occur.

Voltage Build-up in a Shunt DC Generator

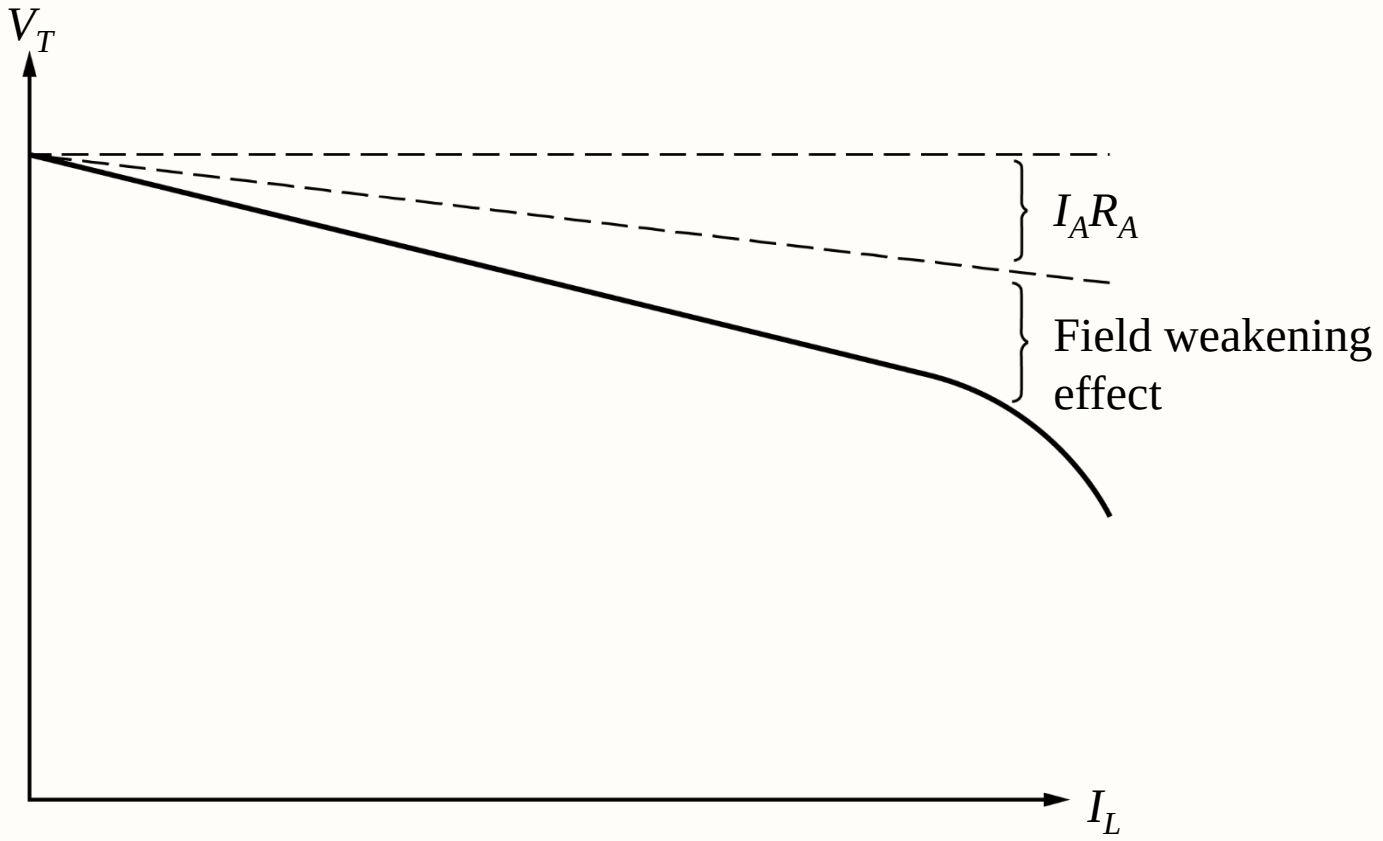
- Normally, the shunt DC generator's voltage will build up to the point where the magnetisation curve intersects the field resistance line, giving the no-load terminal voltage.
- The field resistance should be less than $R_{critical}$ for voltage build-up to occur.
- If R_F exceeds $R_{critical}$, the steady-state operating voltage is essentially at the residual level, and it never builds up.
- The solution to this problem is to reduce R_F.

Voltage Build-up in a Shunt DC Generator



The effect of the field resistance on the no-load terminal voltage.
If $R_F > R_2 = R_{critical}$, the generator's voltage will never build up.

V-I Characteristics of a Shunt DC Generator



When load on the generator $\uparrow \Rightarrow I_L$ (hence I_A) $\uparrow \Rightarrow I_A R_A \uparrow \Rightarrow V_T \downarrow$.

When $V_T \downarrow \Rightarrow I_F \downarrow \Rightarrow \phi \downarrow \Rightarrow E_A \downarrow \Rightarrow V_T \downarrow$ further.

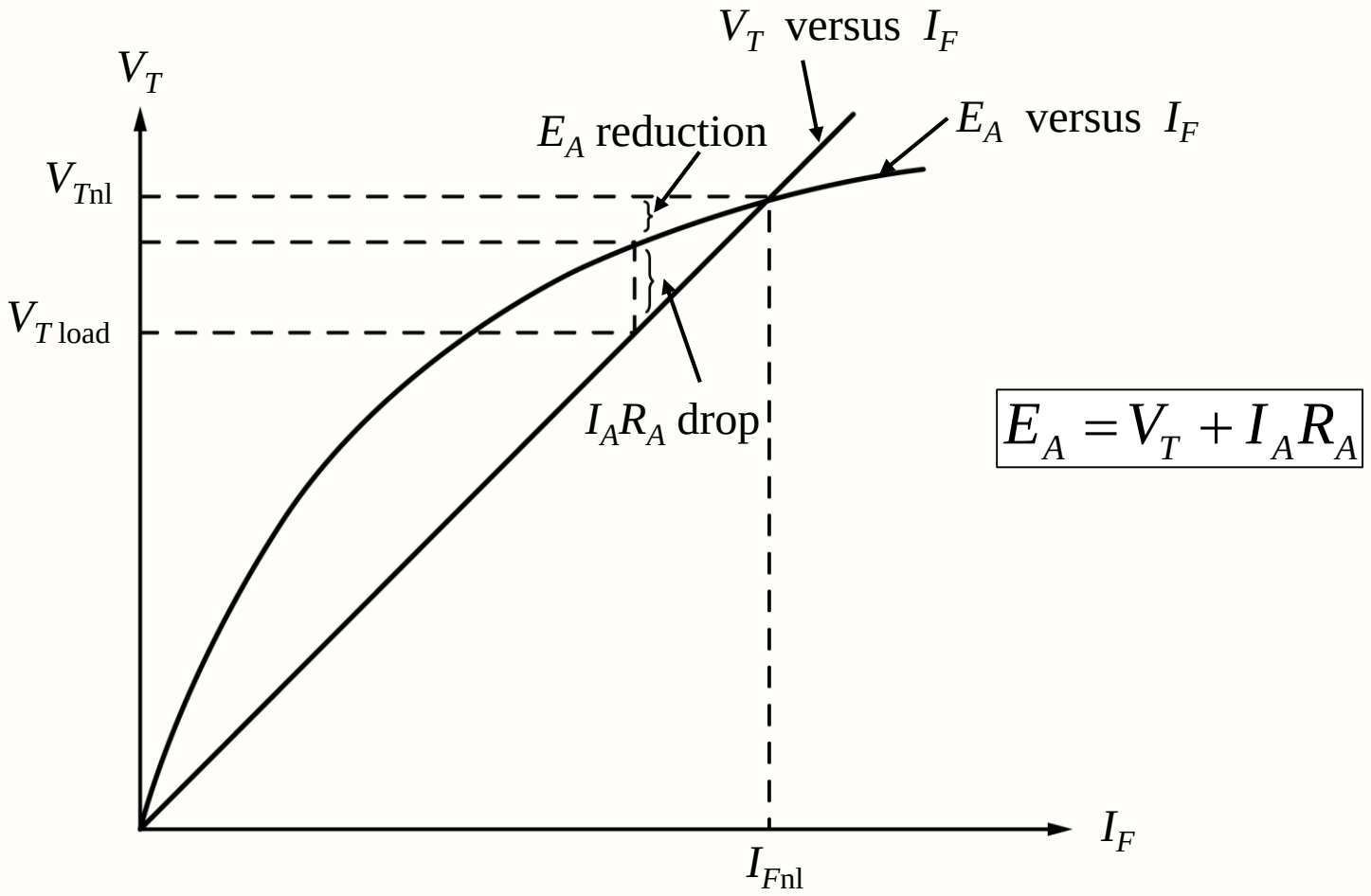
Voltage Control of a Shunt DC Generator

- As with the separately excited DC generator, there are two ways to control the voltage of a shunt DC generator:
  - Change the shaft speed of the generator.
  - Change the total field resistance of the generator, thus changing the field current – this method is widely used.

Analysis of Shunt DC Generators

- The analysis of a shunt DC generator is more complicated than the separately excited generator, because the field current in the machine depends directly on the machine's own output voltage.
- Shunt DC generators with armature reaction will further complicate the analysis, and will not be discussed in this course.

Analysis of Shunt DC Generators



$$E_A = V_T + I_A R_A$$

Graphical analysis of a shunt DC generator with compensating windings.

Analysis of Shunt DC Generators

- The field resistance $R_F = \frac{V_T}{I_F}$ is shown by the straight line laid over the magnetisation curve.
- At no load, $V_T \simeq E_A$ and the generator operates at the voltage where R_F line intersects the magnetisation curve.
- As $E_A = V_T + I_A R_A$, to find the terminal voltage for a given load, just determine the $I_A R_A$ drop and locate the place on the graph, where that drop fits exactly between the E_A line and the V_T line.
- If there are two possible places on the curve where the $I_A R_A$ drop will fit exactly, the one nearer the no-load voltage will represent the normal operating point.

Example 1

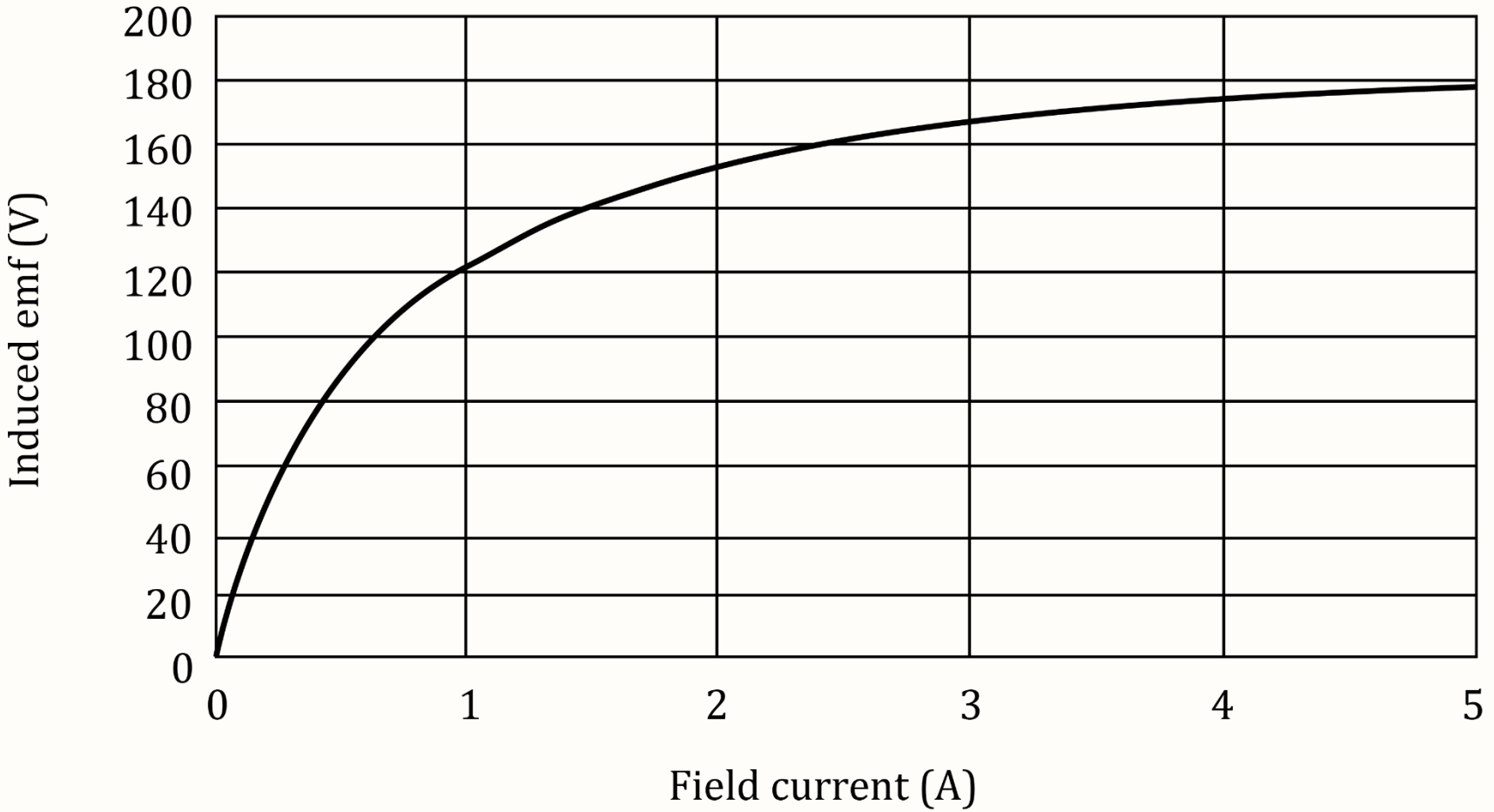
A DC shunt generator is driven at 1200 rpm. The magnetisation curve at 1200 rpm is shown, and the armature and field resistances are 0.2 ohm and 30 ohms, respectively. Neglect armature reaction.

a) An external rheostat is inserted in series with the field winding, and adjusted to give a no-load voltage of 170 V. What is the value of the rheostat setting?

b) If the generator supplies an armature current of 103.5 A, determine the armature terminal voltage with the rheostat setting obtained in (a).

(Solutions →)

Example 1



ELECTRIC MACHINERY AND TRANSFORMERS 3E by Guru et al (2000): Fig 5.36 (p. 327). By permission of Oxford University Press, USA.

Example 1 – Solutions

a) From the magnetisation curve, the field current corresponding to a no-load terminal voltage of 170 V is 3.5 A. The total field-circuit resistance is:

$$R_F = \frac{170}{3.5} = 48.57\Omega$$

Thus, the external rheostat setting in the field-winding circuit is:

$$R_{ext} = 48.57 - 30 = 18.57\Omega$$

Example 1 – Solutions

b) For $I_A = 103.5$ A,

$$V_T = I_F(48.57) = E_A - (103.5)(0.2)$$

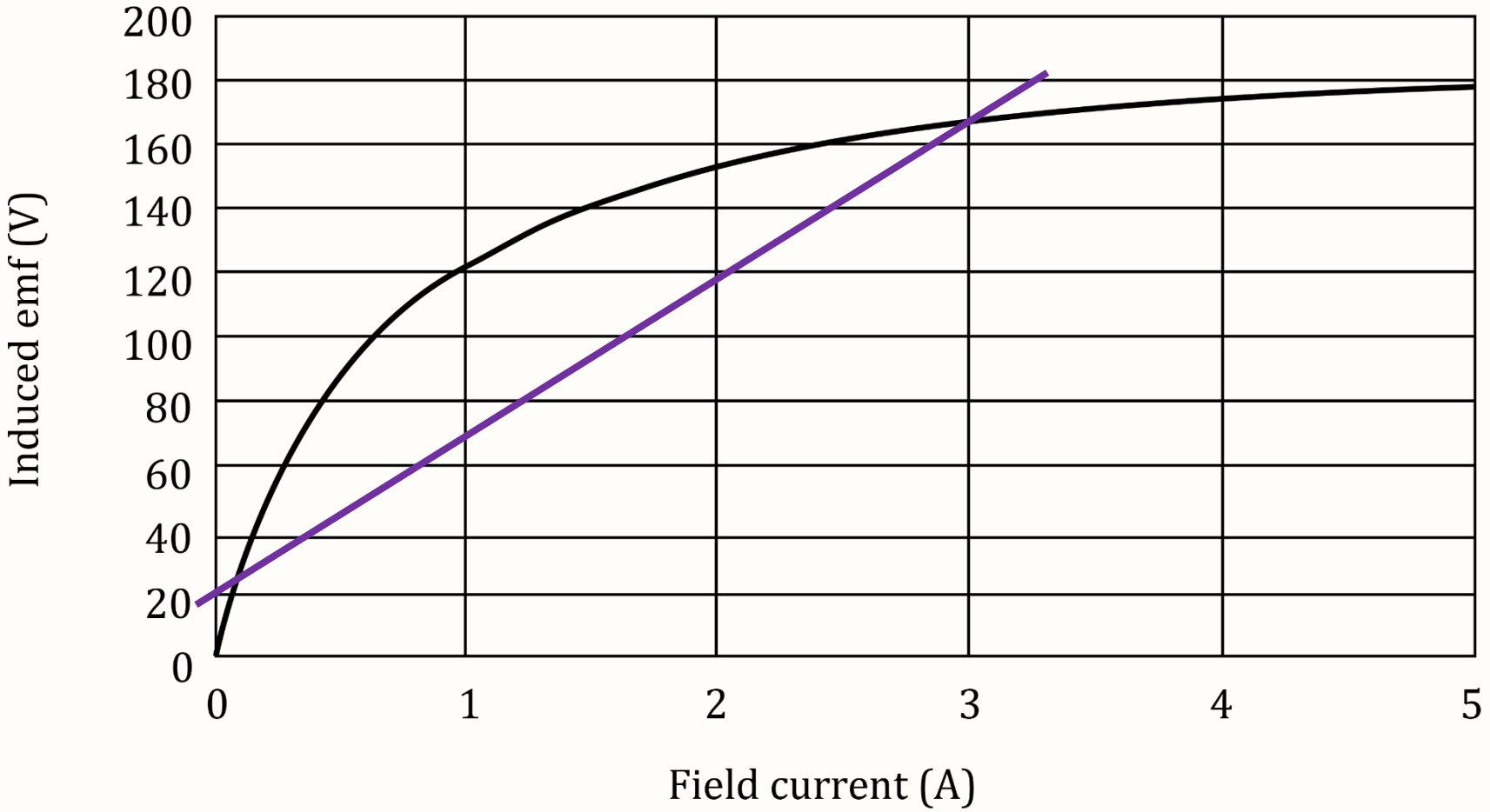
$$\Rightarrow E_A = 48.57 I_F + 20.7$$

Superimposing the above equation onto the magnetisation curve yields the intersection point.

$$I_F = 3 \text{ A}, \; E_A = 166 \text{ V}$$

$$\Rightarrow V_T = 48.57 I_F = 145.71 \text{ V}$$

Example 1 – Solutions



ELECTRIC MACHINERY AND TRANSFORMERS 3E by Guru et al (2000): Fig 5.36 (p. 327). By permission of Oxford University Press, USA.

Losses and Efficiency of a DC Generator

- Consider a shunt DC generator as an example:

Power output, $P_{out} = V_T I_L$

Power input $P_{in}(mech) = T_{in}\omega_m$ where T_{in} is the prime mover's torque.

$$P_{dev} = E_A I_A$$

$$P_{in}(mech) = P_{dev} + P_{rot}$$

Generator efficiency, $\eta = \frac{P_{out}}{P_{in}} \times 100\%$

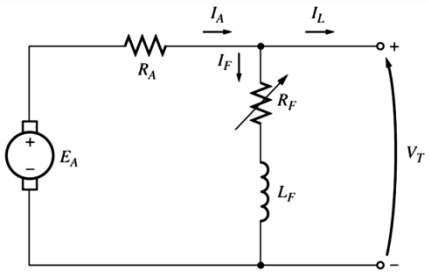
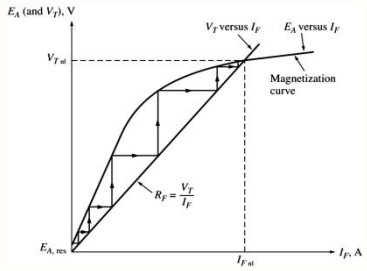
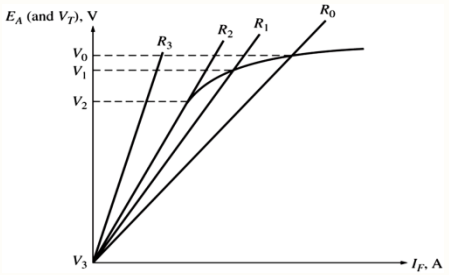
Note: For separately excited DC generator, $P_{in} = T_{in}\omega_m + V_F I_F$

Summary of DC Generators

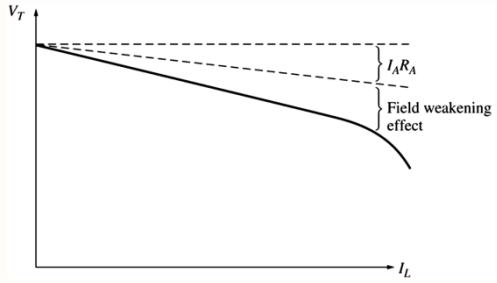
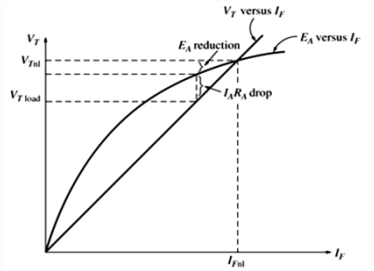
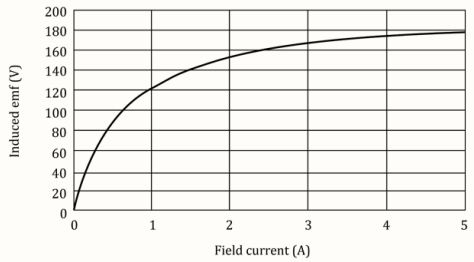
You have learnt:

- DC generators are DC machines used as generators. The different types differ in the manner in which their field fluxes are derived.
- Voltage build-up in self-excited DC generators.
- The voltage-current characteristics of a shunt DC generator.
- The graphical approach to the analysis of shunt DC generator, using the magnetisation curve and the equivalent circuit.

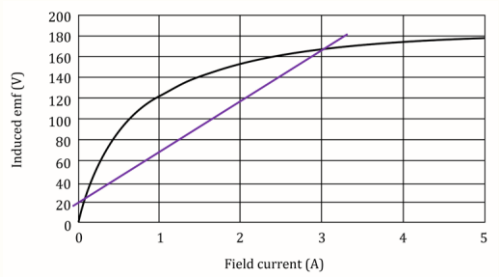
References

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